GeospaceLAB-Swarm Guide

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Contents

0.1 Getting Started

0.1.1 Overview

What is GeospaceLAB?

Research projects in space physics and space weather are typically based on several kinds of measured and/or modelling data. Processing and combining those data are demanding tasks because they are often provided from different data sources and in different formats. Even one satellite mission, such as Swarm, may include tens of data products. The diversity of data adds an unnecessary complexity to the data analysis in research projects. It often takes a lot of time for a researcher to collect and manage the data before the data are processed for a further analysis and interpretation. To improve the productivity in data access and analysis, the open-source Python package GeospaceLAB is developed.

GeospaceLAB provides a unified process for data access, management, and visualization that connects the data provider and the space physics researchers. Using the package, researchers can progress their research in a quick manner and focus more on the data interpretation and research results. The package has been applied in the study of Magnetosphere-ionosphere-thermosphere coupling. The package got 29 stars in GitHub and was included in Python in Heliosphysics Community (PyHC) in 2022.

Supported Swarm data products in GeospaceLAB

As listed in Table 1, GeospaceLAB currently supports 30 data products of Swarm mission. Those data products are selected from the categories of “Ionosphere/Magnetosphere”, “Thermosphere”, “Space Weather”, and “Magnetic measurements” in the Swarm Product Data Handbook.

Users can easily access the listed Swarm data products from (1) ESA’s Swarm Dissemination Sever, (2) Swarm VirES Service*, and (3) VirES for Swarm - HAPI Server*. The data access and visualization are tested and verified in GeospaceLAB.

> **Note**
> *: For VirES and VirES HAPI server, only the MAG data is currently supported. The support for other data products is under development and will be released in the future.

Table 1: Supported Swarm data products in GeospaceLAB

No.	Product	Level	Folder	Description	Status
1	SW_AEJxLPL_2F	2F	Level2daily	Auroral electrojets line profile – Line current method (LC)	Tested
2	SW_AEJxLPS_2F	2F	Level2daily	Auroral electrojets line profile – SECS method	Tested
3	SW_AEJxPBL_2F	2F	Level2daily	Auroral electrojets peaks and boundaries from LC	Tested
4	SW_AEJxPBS_2F	2F	Level2daily	Auroral electrojets peaks and boundaries from SECS	Tested
5	SW_AOBxFAC_2F	2F	Level2daily	Auroral oval boundaries derived from FAC	Tested
6	SW_DNSxACC_2_	2_	Level2daily	Thermospheric density derived from ACC + POD	Tested
7	SW_DNSxPOD_2_	2_	Level2daily	Thermospheric density derived from POD	Tested
8	SW_EFIx_LP_1B	1B	Level1b	Electric field instrument (EFI) Langmuir probe (LP) measurements at 2 Hz	Tested
9	SW_EFIx_LP_FP	Advanced	Advanced/Plasma_Data	EFI faceplate (FP) plasma density at 16 Hz	Tested
10	SW_EFIx_LP_HM	Advanced	Advanced/Plasma_Data	EFI LP extended at 2 Hz	Tested
11	SW_EFIx_TCT02	Advanced	Advanced/Plasma_Data	Extended dataset of EFI TII cross-track ion flow measurements at 2 Hz	Tested
12	SW_EFIx_TCT16	Advanced	Advanced/Plasma_Data	Extended dataset of EFI TII cross-track ion flow measurements at 16 Hz	Tested
13	SW_EFIxIDM_2_	Advanced	Advanced/Plasma_Data	EFI LP ion drift and effec-	Tested

0.1.2 Installation

Python environment and IDEs

The Python distribution **Anaconda** is recommended. To install **Anaconda**, see Installing Anaconda Distribution. After installation, a **conda** virtual environment can be created by, e.g.,

```
conda create - -name YOUR_ENV_NAME python=3.12 cython spyder
```

After creating the virtual environment, one needs to activate it:

```
conda activate YOUR_ENV_NAME
```

Python IDEs Popular Python IDEs include **Spyder**, **PyCharm**, and **VS Code**.

Install GeospaceLAB

Install GeospaceLAB via PyPI:

```
pip install geospacelab
```

To upgrade GeospaceLAB:

```
pip install -U geospacelab
```

Dependencies Most of the dependencies are installed when the package is installed. A few packages can be installed manually.

For example, the **Cartopy** package is useful for processing and mapping geospatial data. To install **Cartopy**, see Installing Cartopy.

Another useful package is **ApexPy**, which provides a Python interface to the Apex coordinate system. To install **ApexPy**, see Installing ApexPy.

0.1.3 Configuration

Initial configuration in GeospaceLAB

When to import GeospaceLAB for the first time, a sequence of messages will be prompted in the Python console and a user can set the root directory of the data to be stored. All of the settings are defined in the configuration file, which can be found here

```
[YOUR_HOME_DIR]/.geospacelab/config.toml
```

The default directory for data storage is

```
[YOUR_HOME_DIR]/Geospacelab/Data
```

If it is not changed, the Swarm data will downloaded from the Swarm Dissemination Sever will be stored in that data folder.

Configuration for Swarm

To dock the Swarm data products, some settings will be made. A message will be prompted in the Python console, when a Swarm dataset is imported at the first time. It asks for the account name and password of ESA Earth Online. Please register the account following the instruction at ESA EO Identity and Access Management System or How do I access Swarm data?.

The account name and password will be stored using Python keyring.

> **Note**
> The Python keyring) package securely stores and retrieves passwords, similar to the MacOS keychain. Using this module, Python code can access the system keyring service. Only users with the correct service name and username can retrieve stored passwords.

> **Tip**
> In case that incorrect account name and/or passward are input, the user can change the account name in the configuration file. Find the following lines and the change the username:
>
> ```
> [datahub.swarm]
> username = "xxx@xxx"
> ```
>
> Then an update of the username and/or passward in keyring is needed. To change the username and password in keyring via Python or command-line, follow this instruction.

Optional: Configuration for Swarm VirES Defaultly GeospaceLAB will download the Swarm data from ESA's Swarm Dissemination Sever and the data will be stored locally. An developing feature is that a VirES user can use GeospaceLAB to touch the Swarm data that supported in VirES and use the functions in GeospaceLAB to manage and visualize the data. Configuration for the VirES Python client can be found in Swarm Notebooks (see Swarm access through VirES)

0.1.4 Quickstart

```
import datetime
import geospacelab.datahub as datahub
import geospacelab.visualization.mpl as gsl_mpl
```

Swarm Data Access

Initial settings

```
dt_fr = datetime.datetime(2016, 3, 15, 18, 18)
dt_to = datetime.datetime(2016, 3, 15, 18, 23)
dt_tsat_id = "A"
```

Create a datahub

```
dh = datahub.DataHub(dt_fr=dt_fr, dt_to=dt_to, visual='on')
```

Dock a Swarm dataset Docking a dataset is the process of loading the dataset into the datahub. It is done by calling the `dock` method of the datahub object. The method takes the name patterns of the dataset as an argument and returns a dataset object.

The following code docks the Swarm MAG LR dataset for the specified time range (shown above) and satellite ID. A full list of the supported datasets and their name patterns can be found in the Supported datasets tutorial.

```
ds = dh.dock(datasource_contents=['esa_eo', 'swarm', 'l1b', 'mag_lr'], sat_id='A', add_APEX=Tr
```

```
Load IGRF coefficients ...

Searching the data product "MAG_LR" with the version "latest" on the server...
The file [PosixPath('/data/afys -ionosphere/data/ESA/SWARM/Level1b/MAG_LR/0701/Sat_A/2016/SW_O
WARNING: Multiple files found for the pattern *20160315T000000*20160315T235959*0701*.cdf in th
/opt/anaconda3/envs/Swarm/lib/python3.12/site -packages/numpy/_core/numeric.py:476: RuntimeWar
  multiarray.copyto(res, fill_value, casting='unsafe')
```

List loaded variables

```
ds.list_all_variables()
```

```
Dataset: esa/earthonline | swarm | mag | mag_lr
Printing all of the variables ...
        No.                   Variable name
         1          SC_DATETIME
         2          SYNC_STATUS
         3          SC_GEO_LAT
         4          SC_GEO_LON
         5          SC_GEO_r
         6          B_VFM
         7          B_VFM_x
         8          B_VFM_y
         9          B_VFM_z
         10         B_VFM_x_err
         11         B_VFM_y_err
         12         B_VFM_z_err
         13         B_NEC
         14         B_N
```

```
15          B_E
16          B_C
17          dB_Sun_VFM
18          dB_Sun_VFM_x
19          dB_Sun_VFM_y
20          dB_Sun_VFM_z
21          dB_AOCS_VFM
22          dB_AOCS_VFM_x
23          dB_AOCS_VFM_y
24          dB_AOCS_VFM_z
25          dB_other_VFM
26          dB_other_VFM_x
27          dB_other_VFM_y
28          dB_other_VFM_z
29          B_VFM_err
30          q_NEC_CRF
31          Att_error
32          FLAG_B
33          FLAG_q
34          FLAG_Platform
35          FLAG_B_BIN_AUX
36          FLAG_B_BIN_IND
37          FLAG_q_BIN_AUX
38          FLAG_q_BIN_IND
39          FLAG_Platform_BIN_AUX
40          FLAG_Platform_BIN_IND
41          FLAG_F
42          FLAG_F_BIN_AUX
43          FLAG_F_BIN_IND
44          ASM_Freq_Dev
45          F
46          F_err
47          dF_Sun
48          dF_AOCS
49          dF_other
50          SC_APEX_LAT
51          SC_APEX_LON
52          SC_APEX_MLT
53          SC_GEO_LST
```

Get a variable and data array

```
B_N = ds['B_N']
B_N
```

```
GeospaceLab Variable object <name: B_N, value: array([[3575.961 ],
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[8864.4029],
[8890.6999],
[8918.0185],
[8945.5857],
[8973.3702],
[8999.8423],
[9026.387 ],
[9053.0208],
[9079.1098],
[9104.7985],
[9131.9597],
[9160.4235],
[9188.5219],
[9215.8336],
[9243.353 ],
[9270.5548],
[9298.3961],
[9326.1167],
[9353.6564],
[9381.2226],
[9409.0971],
[9436.9306],
[9464.6723],
[9492.5535],
[9520.6549],
[9548.5465],
[9576.2863],
[9604.1236],
```
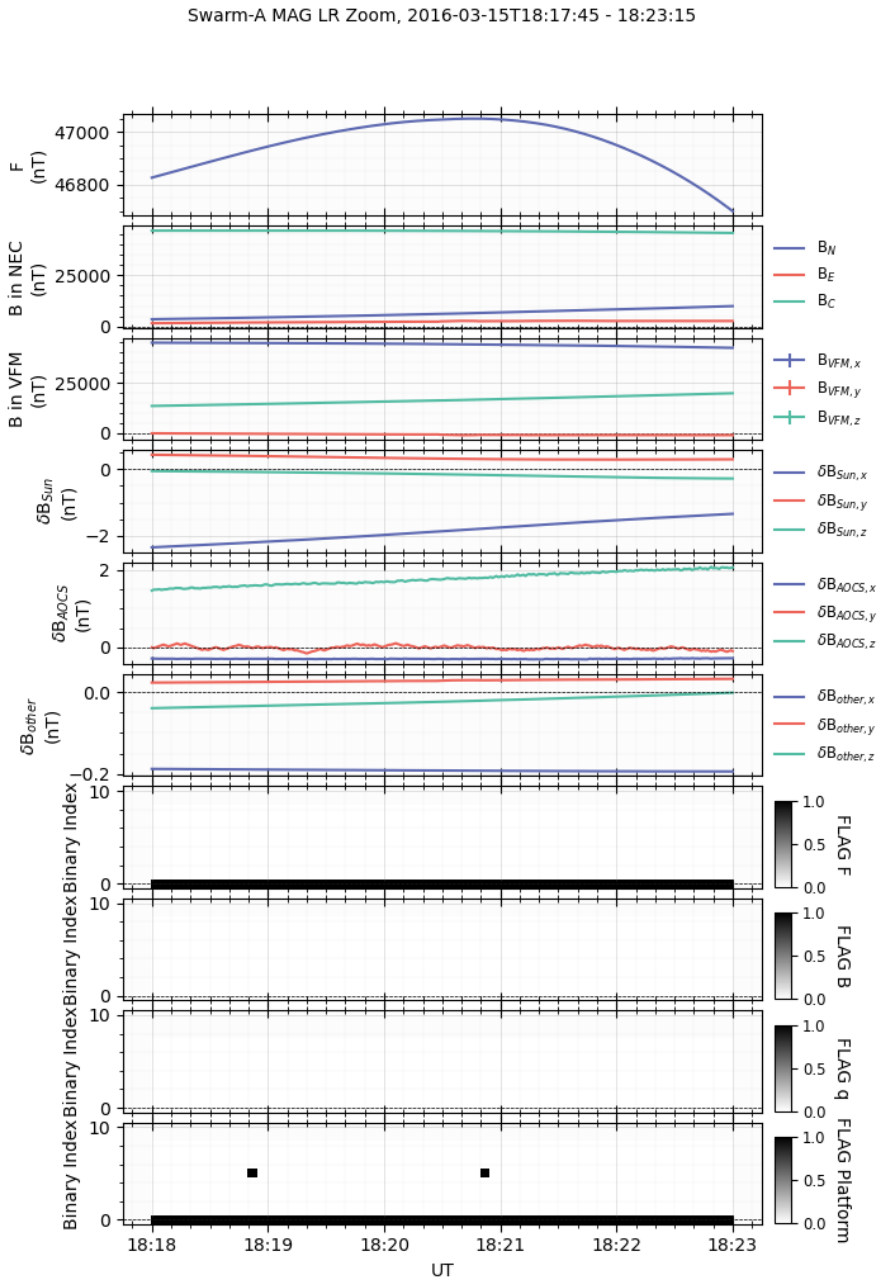

```
       [9632.1933],
       [9660.2264],
       [9688.4443],
       [9717.0215],
       [9745.0572],
       [9773.2622],
       [9801.4568],
       [9829.8285],
       [9858.1734],
       [9886.4872],
       [9914.7995],
       [9943.1433],
       [9971.6779]])
```

Swarm Data Visualization

```
fig = gsl_mpl.create_figure(figsize=(8, 12))
db = gsl_mpl.dashboards.TSDashboard(figure=fig)

panel_layouts = [
    [ds['F']],
        [ds['B_N'], ds['B_E'], ds['B_C']],
        [ds['B_VFM_x'], ds['B_VFM_y'], ds['B_VFM_z']],
        [ds['dB_Sun_VFM_x'], ds['dB_Sun_VFM_y'], ds['dB_Sun_VFM_z']],
        [ds['dB_AOCS_VFM_x'], ds['dB_AOCS_VFM_y'], ds['dB_AOCS_VFM_z']],
        [ds['dB_other_VFM_x'], ds['dB_other_VFM_y'], ds['dB_other_VFM_z']],
        [ds['FLAG_F_BIN_AUX']],
        [ds['FLAG_B_BIN_AUX']],
        [ds['FLAG_q_BIN_AUX']],
        [ds['FLAG_Platform_BIN_AUX']],
    ]
db.set_layout(panel_layouts)
db.draw()
db.add_title(title='Swarm -{} MAG LR Zoom'.format(ds.sat_id), fontsize='medium', append_time=T
db.show()
```


Swarm-A MAG LR Zoom, 2016-03-15T18:17:45 - 18:23:15
F (nT)
B in NEC (nT)
B in VFM (nT)
δB_Sun (nT)
δB_AOCS (nT)
δB_other (nT)
Binary Index
FLAG F
FLAG B
FLAG q
FLAG Platform
UT

0.2 Tutorials

0.2.1 Swarm data access

This tutorial demonstrates how to access Swarm data using GeospaceLAB. We will use the Swarm MAG LR dataset as an example, but the same process applies to other datasets as well. A full list of the supported datasets and their name patterns can be found in the Supported datasets tutorial. Also, the examples for accessing other datasets can be found in the Examples tutorial.

Supported datasets in GeospaceLAB

```
import geospacelab.datahub as datahub
dh = datahub.DataHub()
dh.list_sourced_datasets()
```

```
Load IGRF coefficients ...
data_sources
| - - -CDAWEB
| - - -| - - -DMSP
| - - -| - - -| - - -SSUSI
| - - -| - - -| - - -| - - -EDR_AUR
| - - -| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_content
| - - -| - - -| - - -| - - -| - - -datasource_contents: ['cdaweb', 'dmsp', 'ssusi', 'edr_aur']
| - - -| - - -| - - -| - - -SDR_DISK
| - - -| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_content
| - - -| - - -| - - -| - - -| - - -datasource_contents: ['cdaweb', 'dmsp', 'ssusi', 'sdr_disk'
| - - -| - - -OMNI
| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_contents', 'omni_type
| - - -| - - -| - - -datasource_contents: ['cdaweb', 'omni']
| - - -ESA_EO
| - - -| - - -SWARM
| - - -| - - -| - - -ADVANCED
| - - -| - - -| - - -| - - -EFI_IDM
| - - -| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_content
| - - -| - - -| - - -| - - -| - - -datasource_contents: ['esa_eo', 'swarm', 'advanced', 'efi_i
| - - -| - - -| - - -| - - -EFI_LP_FP
| - - -| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_content
| - - -| - - -| - - -| - - -| - - -datasource_contents: ['esa_eo', 'swarm', 'advanced', 'efi_l
| - - -| - - -| - - -| - - -EFI_LP_HM
| - - -| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_content
| - - -| - - -| - - -| - - -| - - -datasource_contents: ['esa_eo', 'swarm', 'advanced', 'efi_l
| - - -| - - -| - - -| - - -EFI_TCT02
| - - -| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_content
| - - -| - - -| - - -| - - -| - - -datasource_contents: ['esa_eo', 'swarm', 'advanced', 'efi_t
| - - -| - - -| - - -| - - -EFI_TCT16
| - - -| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_content
| - - -| - - -| - - -| - - -| - - -datasource_contents: ['esa_eo', 'swarm', 'advanced', 'efi_t
| - - -| - - -| - - -L1B
| - - -| - - -| - - -| - - -EFI_LP
| - - -| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_content
| - - -| - - -| - - -| - - -| - - -datasource_contents: ['esa_eo', 'swarm', 'l1b', 'efi_lp']
| - - -| - - -| - - -| - - -EFI_LPI
| - - -| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_content
| - - -| - - -| - - -| - - -| - - -datasource_contents: ['esa_eo', 'swarm', 'l1b', 'efi_lpi']
| - - -| - - -| - - -| - - -MAG_HR
| - - -| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_content
| - - -| - - -| - - -| - - -| - - -datasource_contents: ['esa_eo', 'swarm', 'l1b', 'mag_hr']
| - - -| - - -| - - -| - - -MAG_LR
| - - -| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_content
| - - -| - - -| - - -| - - -| - - -datasource_contents: ['esa_eo', 'swarm', 'l1b', 'mag_lr']
| - - -| - - -| - - -L2DAILY
| - - -| - - -| - - -| - - -AEJ_LPL
| - - -| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_content
```

```
| - - -| - - -| - - -| - - -| - - -datasource_contents: ['esa_eo', 'swarm', 'l2daily', 'aej_lp
| - - -| - - -| - - -| - - -AEJ_LPS
| - - -| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_content
| - - -| - - -| - - -| - - -| - - -datasource_contents: ['esa_eo', 'swarm', 'l2daily', 'aej_lp
| - - -| - - -| - - -| - - -AEJ_PBL
| - - -| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_content
| - - -| - - -| - - -| - - -| - - -datasource_contents: ['esa_eo', 'swarm', 'l2daily', 'aej_pb
| - - -| - - -| - - -| - - -AEJ_PBS
| - - -| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_content
| - - -| - - -| - - -| - - -| - - -datasource_contents: ['esa_eo', 'swarm', 'l2daily', 'aej_pb
| - - -| - - -| - - -| - - -AOB_FAC
| - - -| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_content
| - - -| - - -| - - -| - - -| - - -datasource_contents: ['esa_eo', 'swarm', 'l2daily', 'aob_fa
| - - -| - - -| - - -| - - -DNS_ACC
| - - -| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_content
| - - -| - - -| - - -| - - -| - - -datasource_contents: ['esa_eo', 'swarm', 'l2daily', 'dns_ac
| - - -| - - -| - - -| - - -DNS_POD
| - - -| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_content
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| - - -| - - -| - - -| - - -EEF_TMS
| - - -| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_content
| - - -| - - -| - - -| - - -| - - -datasource_contents: ['esa_eo', 'swarm', 'l2daily', 'eef_tm
| - - -| - - -| - - -| - - -EFI_TIE
| - - -| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_content
| - - -| - - -| - - -| - - -| - - -datasource_contents: ['esa_eo', 'swarm', 'l2daily', 'efi_ti
| - - -| - - -| - - -| - - -FAC_LLS_DUAL
| - - -| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_content
| - - -| - - -| - - -| - - -| - - -datasource_contents: ['esa_eo', 'swarm', 'l2daily', 'fac_ll
| - - -| - - -| - - -| - - -FAC_TMS
| - - -| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_content
| - - -| - - -| - - -| - - -| - - -datasource_contents: ['esa_eo', 'swarm', 'l2daily', 'fac_tm
| - - -| - - -| - - -| - - -FAC_TMS_DUAL
| - - -| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_content
| - - -| - - -| - - -| - - -| - - -datasource_contents: ['esa_eo', 'swarm', 'l2daily', 'fac_tm
| - - -| - - -| - - -| - - -IBI_TMS
| - - -| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_content
| - - -| - - -| - - -| - - -| - - -datasource_contents: ['esa_eo', 'swarm', 'l2daily', 'ibi_tm
| - - -| - - -| - - -| - - -IPD_IRR
| - - -| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_content
| - - -| - - -| - - -| - - -| - - -datasource_contents: ['esa_eo', 'swarm', 'l2daily', 'ipd_ir
| - - -| - - -| - - -| - - -MIT_LP
| - - -| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_content
| - - -| - - -| - - -| - - -| - - -datasource_contents: ['esa_eo', 'swarm', 'l2daily', 'mit_lp
| - - -| - - -| - - -| - - -MIT_TEC
| - - -| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_content
| - - -| - - -| - - -| - - -| - - -datasource_contents: ['esa_eo', 'swarm', 'l2daily', 'mit_te
| - - -| - - -| - - -| - - -NIX_TMS
| - - -| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_content
| - - -| - - -| - - -| - - -| - - -datasource_contents: ['esa_eo', 'swarm', 'l2daily', 'nix_tm
| - - -| - - -| - - -| - - -PPI_FAC
| - - -| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_content
| - - -| - - -| - - -| - - -| - - -datasource_contents: ['esa_eo', 'swarm', 'l2daily', 'ppi_fa
| - - -| - - -| - - -| - - -TEC_TMS
```

```
|---|---|---|---|---Required inputs when load_mode="AUTO": ['datasource_content
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|---|---|---|---TIX_TMS
|---|---|---|---|---Required inputs when load_mode="AUTO": ['datasource_content
|---|---|---|---|---datasource_contents: ['esa_eo', 'swarm', 'l2daily', 'tix_tm
|---|---|---|---WHI_EVT
|---|---|---|---|---Required inputs when load_mode="AUTO": ['datasource_content
|---|---|---|---|---datasource_contents: ['esa_eo', 'swarm', 'l2daily', 'whi_ev
|---FMI
|---|---IMAGE
|---|---|---IE
|---|---|---|---Required inputs when load_mode="AUTO": ['datasource_contents']
|---|---|---|---datasource_contents: ['fmi', 'image', 'ie']
|---GFZ
|---|---HPO
|---|---|---Required inputs when load_mode="AUTO": ['datasource_contents']
|---|---|---datasource_contents: ['gfz', 'hpo']
|---|---|---NOWCAST
|---|---|---|---Required inputs when load_mode="AUTO": ['datasource_contents']
|---|---|---|---datasource_contents: ['gfz', 'hpo', 'nowcast']
|---|---KPAP
|---|---|---Required inputs when load_mode="AUTO": ['datasource_contents']
|---|---|---datasource_contents: ['gfz', 'kpap']
|---|---|---NOWCAST
|---|---|---|---Required inputs when load_mode="AUTO": ['datasource_contents']
|---|---|---|---datasource_contents: ['gfz', 'kpap', 'nowcast']
|---|---SNF107
|---|---|---Required inputs when load_mode="AUTO": ['datasource_contents']
|---|---|---datasource_contents: ['gfz', 'snf107']
|---ISEE
|---|---GNSS
|---|---|---TECMAP
|---|---|---|---Required inputs when load_mode="AUTO": ['datasource_contents']
|---|---|---|---datasource_contents: ['isee', 'gnss', 'tecmap']
|---JHUAPL
|---|---AMPERE
|---|---|---FITTED
|---|---|---|---Required inputs when load_mode="AUTO": ['datasource_contents']
|---|---|---|---datasource_contents: ['jhuapl', 'ampere', 'fitted']
|---|---|---GRD
|---|---|---|---Required inputs when load_mode="AUTO": ['datasource_contents']
|---|---|---|---datasource_contents: ['jhuapl', 'ampere', 'grd']
|---|---DMSP
|---|---|---SSUSI
|---|---|---|---EDRAUR
|---|---|---|---|---Required inputs when load_mode="AUTO": ['datasource_content
|---|---|---|---|---datasource_contents: ['jhuapl', 'dmsp', 'ssusi', 'edraur']
|---|---|---|---SDRDISK
|---|---|---|---|---Required inputs when load_mode="AUTO": ['datasource_content
|---|---|---|---|---datasource_contents: ['jhuapl', 'dmsp', 'ssusi', 'sdrdisk']
|---MADRIGAL
|---|---GNSS
|---|---|---TECMAP
```

```
| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_contents']
| - - -| - - -| - - -| - - -datasource_contents: ['madrigal', 'gnss', 'tecmap']
| - - -| - - -ISR
| - - -| - - -| - - -EISCAT
| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_contents', 'si
| - - -| - - -| - - -| - - -datasource_contents: ['madrigal', 'isr', 'eiscat']
| - - -| - - -| - - -MILLSTONEHILL
| - - -| - - -| - - -| - - -BASIC
| - - -| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_content
| - - -| - - -| - - -| - - -| - - -datasource_contents: ['madrigal', 'isr', 'millstonehill', '
| - - -| - - -| - - -| - - -GRIDDED
| - - -| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_content
| - - -| - - -| - - -| - - -| - - -datasource_contents: ['madrigal', 'isr', 'millstonehill', '
| - - -| - - -| - - -| - - -VI
| - - -| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_content
| - - -| - - -| - - -| - - -| - - -datasource_contents: ['madrigal', 'isr', 'millstonehill', '
| - - -| - - -| - - -PFISR
| - - -| - - -| - - -| - - -FITTED
| - - -| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_content
| - - -| - - -| - - -| - - -| - - -datasource_contents: ['madrigal', 'isr', 'pfisr', 'fitted']
| - - -| - - -| - - -| - - -VI
| - - -| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_content
| - - -| - - -| - - -| - - -| - - -datasource_contents: ['madrigal', 'isr', 'pfisr', 'vi']
| - - -| - - -| - - -RISR_N
| - - -| - - -| - - -| - - -FITTED
| - - -| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_content
| - - -| - - -| - - -| - - -| - - -datasource_contents: ['madrigal', 'isr', 'risr_n', 'fitted'
| - - -| - - -| - - -| - - -VI
| - - -| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_content
| - - -| - - -| - - -| - - -| - - -datasource_contents: ['madrigal', 'isr', 'risr_n', 'vi']
| - - -| - - -SATELLITES
| - - -| - - -| - - -DMSP
| - - -| - - -| - - -| - - -E
| - - -| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_content
| - - -| - - -| - - -| - - -| - - -datasource_contents: ['madrigal', 'satellites', 'dmsp', 'e'
| - - -| - - -| - - -| - - -S1
| - - -| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_content
| - - -| - - -| - - -| - - -| - - -datasource_contents: ['madrigal', 'satellites', 'dmsp', 's1
| - - -| - - -| - - -| - - -S4
| - - -| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_content
| - - -| - - -| - - -| - - -| - - -datasource_contents: ['madrigal', 'satellites', 'dmsp', 's4
| - - -NCEI
| - - -| - - -DMSP
| - - -| - - -| - - -SSM_MFR
| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_contents', 'sa
| - - -| - - -| - - -| - - -datasource_contents: ['ncei', 'dmsp', 'ssm_mfr']
| - - -NIPR
| - - -| - - -ASC
| - - -| - - -| - - -TRO_WMI
| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_contents', 'si
| - - -| - - -| - - -| - - -datasource_contents: ['nipr', 'asc', 'tro_wmi']
| - - -SUPERDARN
| - - -| - - -POTMAP
```

```
| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_contents']
| - - -| - - -| - - -datasource_contents: ['superdarn', 'potmap']
| - - -SUPERMAG
| - - -| - - -INDICES
| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_contents']
| - - -| - - -| - - -datasource_contents: ['supermag', 'indices']
| - - -| - - -MAGNETOMETER
| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_contents']
| - - -| - - -| - - -datasource_contents: ['supermag', 'magnetometer']
| - - -TUD
| - - -| - - -CHAMP
| - - -| - - -| - - -DNS_ACC
| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_contents']
| - - -| - - -| - - -| - - -datasource_contents: ['tud', 'champ', 'dns_acc']
| - - -| - - -| - - -WND_ACC
| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_contents']
| - - -| - - -| - - -| - - -datasource_contents: ['tud', 'champ', 'wnd_acc']
| - - -| - - -GOCE
| - - -| - - -| - - -DNS_ACC
| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_contents']
| - - -| - - -| - - -| - - -datasource_contents: ['tud', 'goce', 'dns_acc']
| - - -| - - -| - - -DNS_WND_ACC_V01
| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_contents']
| - - -| - - -| - - -| - - -datasource_contents: ['tud', 'goce', 'dns_wnd_acc_v01']
| - - -| - - -| - - -WND_ACC
| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_contents', 'sa
| - - -| - - -| - - -| - - -datasource_contents: ['tud', 'goce', 'wnd_acc']
| - - -| - - -GRACE
| - - -| - - -| - - -DNS_ACC
| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_contents', 'sa
| - - -| - - -| - - -| - - -datasource_contents: ['tud', 'grace', 'dns_acc']
| - - -| - - -| - - -WND_ACC
| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_contents', 'sa
| - - -| - - -| - - -| - - -datasource_contents: ['tud', 'grace', 'wnd_acc']
| - - -| - - -GRACE_FO
| - - -| - - -| - - -DNS_ACC
| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_contents', 'sa
| - - -| - - -| - - -| - - -datasource_contents: ['tud', 'grace_fo', 'dns_acc']
| - - -| - - -| - - -WND_ACC
| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_contents', 'sa
| - - -| - - -| - - -| - - -datasource_contents: ['tud', 'grace_fo', 'wnd_acc']
| - - -| - - -SWARM
| - - -| - - -| - - -DNS_ACC
| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_contents', 'sa
| - - -| - - -| - - -| - - -datasource_contents: ['tud', 'swarm', 'dns_acc']
| - - -| - - -| - - -DNS_POD
| - - -| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_contents', 'sa
| - - -| - - -| - - -| - - -datasource_contents: ['tud', 'swarm', 'dns_pod']
| - - -WDC
| - - -| - - -AE
| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_contents']
| - - -| - - -| - - -datasource_contents: ['wdc', 'ae']
| - - -| - - -ASYSYM
```

```
| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_contents']
| - - -| - - -| - - -datasource_contents: ['wdc', 'asysym']
| - - -| - - -DST
| - - -| - - -| - - -Required inputs when load_mode="AUTO": ['datasource_contents']
| - - -| - - -| - - -datasource_contents: ['wdc', 'dst']
```

0.2.2 Visualize Swarm data

0.2.3 Data grouping and gridding for LEO satellite data

0.2.4 Conjuction finder

0.2.5 Examples

Swarm MAG data

Swarm EFI data